# 1 Basic Stochastic Convex Optimization: Rates and Lower Bounds for SGD

## 1.1 Setup and Noise Assumptions

$f$ is convex and $L$-smooth, $S := \arg\min f \neq \emptyset$, $x^\star \in S$, $D := \|x_0 - x^\star\|$, and $\Delta_t := f(x_t) - f^\star$. SGD (Robbins and Monro, 1951) is given by

$$x_{t+1} = x_t - s_t\big(\nabla f(x_t) + e_t\big),$$

where $\mathbb{E}_t[\cdot] := \mathbb{E}[\cdot \mid \mathcal{F}_t]$ and $e_t$ is the stochastic gradient error. The literature uses different noise assumptions. We refer to them below by the labels in Table 1.

Table 1: Noise assumptions ($g_k = \nabla f(\cdot) + e_k$ is a stochastic gradient)

| Label | Name | Mathematical form | References |
|---|---|---|---|
| (N1) | Conditional unbiasedness | $\mathbb{E}[e_k \mid \mathcal{F}_k] = 0$ | All works |
| (N2) | Bounded variance | $\mathbb{E}[\|e_k\|^2 \mid \mathcal{F}_k] \le \sigma^2$ | Lan (2012); JOTA setting |
| (N3) | Bounded gradients | $\|\partial f(x)\| \le G$ or $\|g(x,\xi)\| \le G$ | Nemirovski et al. (2009) (smooth) |
| (N4) | ABC condition | $\mathbb{E}\|g_k\|^2 \le A\,(f(x_k) - f^\star) + B\,\|\nabla f(x_k)\|^2 + C$ | Liu and Yuan (2022) |
| (N5) | Relative noise | $\mathbb{E}[\|e_k\|^2 \mid \mathcal{F}_k] \le \nu^2 \|\nabla f(y_k)\|^2$ | Vaswani et al. (2019); Gupta et al. (2024); Hermant et al. (2025) |
| (N6) | Decaying noise | $\sigma_k^2 := \mathbb{E}[\|e_k\|^2 \mid \mathcal{F}_k]$, $(s_k t_k \sigma_k) \in \ell^2$ or $\ell^1$ | Attouch et al. (2026) |

(N2) and (N3) are special cases of (N4). We state (N5) at the gradient evaluation point $y_k$, since stochastic Nesterov evaluates the gradient at the extrapolated point.

## 1.2 Basic Stochastic Convex Optimization: Rates and Lower Bounds for SGD

Under (N1), (N2) and $s_t \le 1/(2L)$, expand $\|x_{t+1} - x^\star\|^2$ and take conditional expectations (see Appendix A):

$$\mathbb{E}_t\|x_{t+1} - x^\star\|^2 + s_t\Delta_t \le \|x_t - x^\star\|^2 + s_t^2\sigma^2. \tag{1}$$

**Interpretation:** the distance decreases by at least $s_t\Delta_t$, at the cost of the noise term $s_t^2\sigma^2$.

Let $S_k := \sum_{t<k} s_t$ and $\widetilde{x}_k := S_k^{-1}\sum_{t<k} s_t x_t$. Taking total expectations in (1), summing from 0 to $k-1$, and applying Jensen's inequality gives:

$$\boxed{\mathbb{E}[f(\widetilde{x}_k) - f^\star] \le \underbrace{\frac{D^2}{S_k}}_{\text{bias}} + \underbrace{\frac{\sigma^2\sum_{t<k} s_t^2}{S_k}}_{\text{variance}}.} \tag{2}$$

**Interpretation:** the bias term decreases as the step size grows, the variance term increases with the step size, and the rate comes from balancing the two.

#### 1.2.1 Rates from (2)

| Step size | Expected rate (weighted average) | Description |
|---|---|---|
| constant $s \asymp \min\{1/L,\, D/(\sigma\sqrt{T})\}$, with known $T$ | $O\big(LD^2/T + \sigma D/\sqrt{T}\big)$ | balancing $D^2/(sT)$ and $s\sigma^2$ (Nemirovski et al., 2009) |
| $s_t \asymp t^{-p}$, $p < 1/2$ | $O(k^{-p})$ | variance |
| $p = 1/2$ | $O(\log k/\sqrt{k})$ | best choice when $T$ is unknown |
| $p = 1/2 + \varepsilon$, $\varepsilon \in (0, 1/2)$ | $O(k^{-1/2+\varepsilon})$ | $\sum_t s_t^2 < \infty$; arbitrarily close to the critical $k^{-1/2}$ rate |

The last three rows follow by directly substituting $s_t \asymp t^{-p}$ into (2).

#### 1.2.2 Lower Bound Including the Bias Term

Under the smoothness assumption and (N1), (N2), the minimax lower bound in the stochastic first-order oracle model is

$$\Omega\Big(\frac{LD^2}{T^2} + \frac{\sigma D}{\sqrt{T}}\Big).$$

The lower bound for the bias term follows from deterministic first-order methods (Nesterov, 2004, Theorem 2.1.7); the lower bound for the noise term follows from information-theoretic arguments (Nemirovski and Yudin, 1983; Agarwal et al., 2012). With the AC-SA method, both terms are attained simultaneously (Lan, 2012). Thus, SGD is already optimal in the noise term; acceleration only improves the bias term from $LD^2/T$ to $LD^2/T^2$. As $k \to \infty$, the asymptotic order is determined by $\sigma D/\sqrt{k}$: **there is no asymptotic acceleration under persistent noise (N2).**

### 1.3 Some Remarks on Almost-sure Analysis

The basic sufficient conditions are

$$\sum_t s_t = \infty, \qquad \sum_t s_t^2 < \infty.$$

Under these conditions, the noise term in (1) is summable, and the Robbins–Siegmund lemma (Robbins and Siegmund, 1971) gives

$$\sum_t s_t \Delta_t < \infty \qquad \text{a.s.}$$

For polynomial stepsizes $s_t \asymp t^{-p}$, square summability requires $p > 1/2$. Writing $p = 1/2 + \varepsilon$ with arbitrarily small $\varepsilon > 0$, the expectation bound above becomes $O(k^{-1/2+\varepsilon})$. Thus, within this stepsize-based analysis, the rate can approach but does not attain the critical $k^{-1/2}$ rate; a corresponding almost-sure rate requires an additional argument beyond the summability conclusion above.

## 2 Stochastic Nesterov: The Same Structure and the Same Limit

Stochastic Nesterov uses

$$y_k = x_k + \alpha_k(x_k - x_{k-1}), \qquad x_{k+1} = y_k - s_k(\nabla f(y_k) + e_k).$$

The momentum sequence is encoded by

$$t_k := 1 + \sum_{i \geq k} \prod_{j=k}^{i} \alpha_j, \qquad \text{equivalently} \qquad \alpha_k = \frac{t_k - 1}{t_{k+1}},$$

and we set $z_k := x_{k-1} + t_k(x_k - x_{k-1})$. JOTA (Attouch et al., 2026, Theorem 3.1) assumes that $f$ is convex, $\nabla f$ is $L$-Lipschitz, $S \neq \emptyset$, $(s_k) \subset (0, 1/L]$ is nonincreasing, $x_0 = x_1$, $\mathcal{F}_k := \sigma(x_i, e_{i-1} : i \leq k)$ and $\sigma_k^2 := \mathbb{E}[\|e_k\|^2 \mid \mathcal{F}_k]$. The conditions on the momentum and the noise are:

$(K_0)$ $\sum_{i \geq k} \prod_{j=k}^{i} \alpha_j < \infty$ for all $k \geq 1$, i.e. $t_k$ is finite.

$(K_1)$ $t_{k+1}^2 - t_k^2 \leq t_{k+1}$ for all $k \geq 1$. This bounds the growth of $t_k$ (it implies $t_{k+1} \leq 2t_k$) and makes the function-value term in the one-step inequality below nonpositive. For $\alpha_k = 1 - \alpha/k$ with $\alpha > 1$, one has $t_{k+1} = k/(\alpha - 1)$, and $(K_1)$ holds if and only if $\alpha \geq 3$.

$(K_1^+)$ There is $m \in [0, 1)$ such that $t_{k+1}^2 - t_k^2 \leq m\, t_{k+1}$ for all $k \geq 1$. This leaves a strictly negative margin in the one-step inequality, which gives summability and $o$ rates. For $\alpha_k = 1 - \alpha/k$, $(K_1^+)$ holds if and only if $\alpha > 3$ (with $m = 2/(\alpha - 1)$).

$(K_2)$ $\mathbb{E}[e_k \mid \mathcal{F}_k] = 0$ and $\sum_k s_k^2 t_k^2 \sigma_k^2 < \infty$ a.s., i.e. $(s_k t_k \sigma_k) \in \ell_+^2(\mathcal{F})$. This makes the noise term in the one-step inequality summable.

$(K_2^+)$ $\mathbb{E}[e_k \mid \mathcal{F}_k] = 0$ and $\sum_k s_k t_k \sigma_k < \infty$ a.s., i.e. $(s_k t_k \sigma_k) \in \ell_+^1(\mathcal{F})$. This is used for iterate convergence and for the transfer to the Ravine sequence $y_k$.

The Lyapunov energy used in JOTA is

$$V_k = s_k\, t_k^2 (f(x_k) - f^\star) + \tfrac{1}{2} \operatorname{dist}(z_k, S)^2.$$

Using the unbiasedness in $(K_2)$, one step gives (Attouch et al., 2026, eq. (18))

$$\mathbb{E}[V_{k+1} \mid \mathcal{F}_k] \leq V_k - s_k\big(t_k^2 - t_{k+1}^2 + t_{k+1}\big)(f(x_k) - f^\star) + s_k^2 t_{k+1}^2 \sigma_k^2.$$

Under $(K_1)$, the middle term is nonpositive and $t_{k+1} \leq 2t_k$, so

$$\mathbb{E}[V_{k+1} \mid \mathcal{F}_k] \leq V_k + 4 s_k^2 t_k^2 \sigma_k^2.$$

Under $(K_1^+)$, the middle term is at most $-(1-m) s_k t_{k+1} (f(x_k) - f^\star)$ and is kept. Under the summability in $(K_2)$, the Robbins–Siegmund lemma gives that $V_k$ converges a.s., hence $f(x_k) - f^\star = O(1/(s_k t_k^2))$ a.s. Thus the noise enters only through the weighted square term. Taking expectations and summing yields (Attouch et al., 2026, Theorem 3.1(i))

$$\mathbb{E}[f(x_k) - f^\star] \leq \underbrace{\frac{\mathbb{E}[V_1]}{s_k t_k^2}}_{\text{bias}} + \underbrace{\frac{4 \sum_{i \leq k} s_i^2 t_i^2\, \mathbb{E}[\sigma_i^2]}{s_k t_k^2}}_{\text{variance}}, \qquad \mathbb{E}[V_1] = s_1 t_1^2\, \mathbb{E}[f(x_0) - f^\star] + \tfrac{1}{2}\mathbb{E}\big[\operatorname{dist}(x_0, S)^2\big].$$

This is the same bias–variance structure as (2), with $s_k t_k^2$ replacing $\sum_{t<k} s_t$. Acceleration enlarges the denominator, but it also amplifies the noise by $t_i^2$.

For $t_k \asymp k$, the two principal regimes are:

| Noise regime | Parameters | Bias | Variance | Consequence |
|---|---|---|---|---|
| Decaying noise (N6) | $s$ constant, $\sum_k k^2\sigma_k^2 < \infty$ | $1/k^2$ | $1/k^2$ | acceleration is preserved; with $(K_1^+)$, a.s. $o(1/k^2)$ |
| Persistent noise (N2) | $\sigma_k \equiv \sigma$, $s_k = ck^{-\delta}$ | $k^{\delta-2}$ | $k^{1-\delta}$ | balance at $\delta = 3/2$, giving expected $O(\log k/\sqrt{k})$ |

Under persistent noise, almost-sure analysis requires $\sum_k k^{2-2\delta} < \infty$, hence $\delta > 3/2$. The resulting rates are slightly slower than $k^{-1/2}$ and can approach it arbitrarily closely. Therefore stochastic Nesterov and SGD have the same noise-dominated order under persistent noise; acceleration survives when the noise decays.

## 3 Research Landscape of Stochastic Acceleration

Results for stochastic acceleration must be compared under the same noise model, evaluation point, convergence notion, and total sample budget. Whether stochastic Nesterov accelerates depends on the noise model; Table 2 summarizes the three cases.

Table 2: Does stochastic Nesterov accelerate? Summary by noise model

| Noise | Acceleration | Known results | Open |
|---|---|---|---|
| (N2) persistent | No | The minimax lower bound fixes the order at $\sigma D/\sqrt{T}$, attained by AC-SA (Nemirovski and Yudin, 1983; Agarwal et al., 2012; Lan, 2012); a.s. rates of momentum methods approach $k^{-1/2}$ at best (Sebbouh et al., 2021; Liu and Yuan, 2022; Attouch et al., 2026); the iterates converge weakly a.s. for all $(t_k)$ satisfying $(K_1)$ when $(\gamma_k t_k) \in \ell^2$ and $\gamma_k t_{k-1}^2 \to \infty$ (Salzo, 2025). | Iterate convergence with state-dependent variance |
| (N5) relative | Yes | Expected $O(1/k^2)$ under suitable noise ratios (Vaswani et al., 2019; Gupta et al., 2024; Mishkin et al., 2024); a.s. $o(1/k^2)$ for a two-step-size SNAG under strong growth, for every $\rho \geq 1$ (Hermant et al., 2025); standard parameters may fail to accelerate (Kidambi et al., 2018; Liu and Belkin, 2020). | Iterate convergence; a.s. rates of the single-step-size scheme |
| (N6) decaying | Yes, if the noise decays fast enough | General-$\alpha_k$ SNAG/SRAG: a.s. $O(1/(s_k t_k^2))$, $o$ rates, summable gradients, and iterate convergence (Attouch et al., 2026; Salzo, 2025). | $\ell^2$ condition for adaptive $\sigma_k$ |

Under persistent noise, acceleration improves only the bias term, so asymptotic acceleration is ruled out. The iterates converge almost surely for every extrapolation sequence satisfying $(K_1)$, with vanishing step sizes and $(\gamma_k t_k) \in \ell^2$ (Salzo, 2025). Decaying noise is essentially covered by JOTA; for uniformly bounded variance, Salzo's $\ell^2$ condition on $(\gamma_k t_k)$ replaces the $\ell^1$ condition $(K_2^+)$. Under relative noise, acceleration is real: expected $O(1/k^2)$ holds under suitable noise ratios, and a two-step-size SNAG attains a.s. $o(1/k^2)$ for every strong growth constant (Hermant et al., 2025). Iterate convergence, and the single-step-size scheme with $\alpha_k = 1 - \alpha/k$, remain open.

# 4 JOTA and the Remaining Questions

JOTA studies the regime of general $\alpha_k \times$ decaying noise $\times$ SNAG/SRAG. It extends momentum beyond constant or step-size-driven choices, permits nonincreasing step sizes, including $s_k \to 0$, with decaying noise, and transfers the analysis from Nesterov to Ravine through their parameter correspondence.

Table 3: Known results and open gaps for $\alpha_k = 1 - \alpha/k$

| Regime | Known results | Open gap | JOTA |
|---|---|---|---|
| $\alpha = 3$, decaying noise (N6) | a.s. $O(1/(s_k k^2))$ under $(K_1)$–$(K_2)$ (Attouch et al., 2026); weak a.s. iterate convergence (Salzo, 2025) | a.s. $o(1/k^2)$ | $O$ rate only |
| $\alpha > 3$, decaying noise (N6) | a.s. $o(1/(s_k k^2))$, summable gradients, and iterate convergence under $(K_2^+)$ (Attouch et al., 2026); $\ell^2$ condition for bounded variance (Salzo, 2025) | $\ell^2$ condition for adaptive $\sigma_k$ | covered |
| $\alpha \geq 3$, persistent noise (N2) | rates arbitrarily close to $k^{-1/2}$ (Attouch et al., 2026; Sebbouh et al., 2021; Liu and Yuan, 2022); weak a.s. iterate convergence with vanishing steps (Salzo, 2025) | iterate convergence with state-dependent variance | rate only |
| $\alpha \geq 3$, relative noise (N5) | expected $O(1/k^2)$ under suitable ratios (Vaswani et al., 2019; Gupta et al., 2024); a.s. $o(1/k^2)$ for the two-step-size SNAG (Hermant et al., 2025) | iterate convergence; a.s. rates of the single-step-size scheme | open |

Three questions remain. First, under relative noise and $\alpha_k = 1 - \alpha/k$ with a single step size, do the iterates converge almost surely, and for which noise ratios $\nu_0(\alpha)$ does the a.s. $o(1/k^2)$ rate hold? Gupta et al. cover $\alpha = 3$ with $\nu < 1$ in expectation, and Hermant et al. cover every strong growth constant with a two-step-size scheme. Second, does Salzo's $\ell^2$ condition extend from uniformly bounded variance to adaptive variances $\sigma_k$, which contain relative noise? Third, determine whether an a.s. $o$ rate is possible at the critical value $\alpha = 3$ under decaying noise, where $(K_1^+)$ fails.

# A Derivation of the One-step Inequality

Expanding the squared distance gives

$$\|x_{t+1} - x^\star\|^2 = \|x_t - x^\star\|^2 - 2s_t\langle \nabla f(x_t) + e_t, x_t - x^\star\rangle + s_t^2\|\nabla f(x_t) + e_t\|^2.$$

By (N1), the conditional expectation of the cross term with $e_t$ is zero. Convexity gives $\langle \nabla f(x_t), x_t - x^\star\rangle \geq \Delta_t$, and smooth convexity gives $\|\nabla f(x_t)\|^2 \leq 2L\Delta_t$ (Nesterov, 2004, Theorem 2.1.5). Hence

$$\mathbb{E}_t\|x_{t+1} - x^\star\|^2 \leq \|x_t - x^\star\|^2 - s_t(2 - 2Ls_t)\Delta_t + s_t^2\sigma^2.$$

When $s_t \leq 1/(2L)$, this implies (1).

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